

1. d

7

$J(\theta)$ is convex, twice differentiable, Largest eigen value of its Hessian matrix $H = \nabla^2 J(\theta) \leq \beta_{\max}$ for all points θ .

The Gradient descent equation is

$$\theta^{[t+1]} = \theta^{[t-1]} - \alpha \nabla J(\theta^{[t-1]})$$

plugging in $x = \theta^{[t-1]}$, $y = \theta^{[t]}$, $y - x = -\alpha \nabla J(\theta^{[t-1]})$ in the Taylor's polynomial from the hint, we get

$$J(\theta^{[t+1]}) = J(\theta^{[t-1]}) + \nabla J(\theta^{[t-1]})^T (-\alpha \nabla J(\theta^{[t-1]})) + \frac{1}{2} (-\alpha \nabla J(\theta^{[t-1]}))^T \nabla^2 J(\theta^{[t-1]} + c(\theta^{[t]} - \theta^{[t-1]})) (-\alpha \nabla J(\theta^{[t-1]}))$$

From Taylor's theorem with Lagrange remainder,

$$J(\theta^{[t+1]}) = J(\theta^{[t-1]}) + \nabla J(\theta^{[t-1]})^T (-\alpha \nabla J(\theta^{[t-1]})) + \frac{1}{2} (-\alpha \nabla J(\theta^{[t-1]}))^T H (-\alpha \nabla J(\theta^{[t-1]}))$$

$$\text{where } H = \nabla^2 J(\theta^{[t-1]} + c(\theta^{[t]} - \theta^{[t-1]}))$$

$$J(\theta^{[t+1]}) = J(\theta^{[t-1]}) - \alpha (\nabla J(\theta^{[t-1]}))^T + \frac{\alpha^2}{2} \nabla J(\theta^{[t-1]})^T H \nabla J(\theta^{[t-1]})$$

Hessian is a symmetric matrix, the maximum value of its quadratic form $(v^T H v)$ is the maximum value of its eigen vector. where v is in its unit form.

$$\lambda_{\max}(H) = \max(v^T H v) \rightarrow \text{unit vector.}$$

$$\frac{\nabla J(\theta^{[t-1]})^T H \nabla J(\theta^{[t-1]})}{\|\nabla J(\theta^{[t-1]})\|^2} \leq \beta_{\max}.$$

1. d

8

$$\nabla J(\theta^{[t-1]})^T H \nabla J(\theta^{[t-1]}) \leq \beta_{\max} \|\nabla J(\theta^{[t-1]})\|^2$$

therefore plugging above in earlier equation, we get below inequality

$$J(\theta^{[t]}) \leq J(\theta^{[t-1]}) + \left(\frac{\beta_{\max}}{2} - \alpha \right) \|\nabla J(\theta^{[t-1]})\|^2$$

$$J(\theta^{[t]}) \leq J(\theta^{[t-1]}) - \alpha \left(1 - \frac{\beta_{\max}}{2} \right) \|\nabla J(\theta^{[t-1]})\|^2$$

this ⁱⁿ equation is ~~an~~ ^{important} iterative equation for gradient descent

For $J(\theta^{[t]})$ to be strictly decreasing

$J(\theta^{[t]})$ should be $< J(\theta^{[t-1]})$
for any value of t .

Since Given $0 < \alpha < 1/\beta_{\max}$

$$\alpha < \frac{1}{\beta_{\max}}$$

$$\alpha \beta_{\max} < 1 \Rightarrow -\frac{\alpha \beta_{\max}}{2} > -\frac{1}{2}$$

$$\Rightarrow 1 - \frac{\alpha \beta_{\max}}{2} > \frac{1}{2} \text{ i.e. } 1 - \frac{\alpha \beta_{\max}}{2} > 0$$

positive value

i.e. to compute $J(\theta^{[t]})$, we are subtracting a positive value from $J(\theta^{[t-1]})$

i.e. $J(\theta^{[t]}) = J(\theta^{[t-1]}) - \text{positive value}$
for all t 's.

Hence, we can say $J(\theta)$ is a decreasing function

1.1 Now coming to convergence proof,

9

for a strictly decreasing function, so long as we are taking on smaller values of α , we should definitely converge in an iterative implementation with trail and error.